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DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY - IST

Assignment Report on

Implementation of Virtualization and Virtual Machines

in

**DISTRIBUTED SYSTEMS AND COMPUTING –
IT23601**

Submitted by

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Implementation of Virtualization and Virtual Machines using VMware Workstation

1. Introduction

Virtualization is a technology that allows multiple operating systems to run on a single physical machine by creating virtual versions of hardware resources. A Virtual Machine (VM) is a software-based computer that behaves like a physical system.

In this experiment, virtualization was implemented using VMware Workstation 17. An Ubuntu Linux operating system was installed as a virtual machine inside a Windows 11 host system.

2. Objective of the Experiment

The objectives of this experiment are:

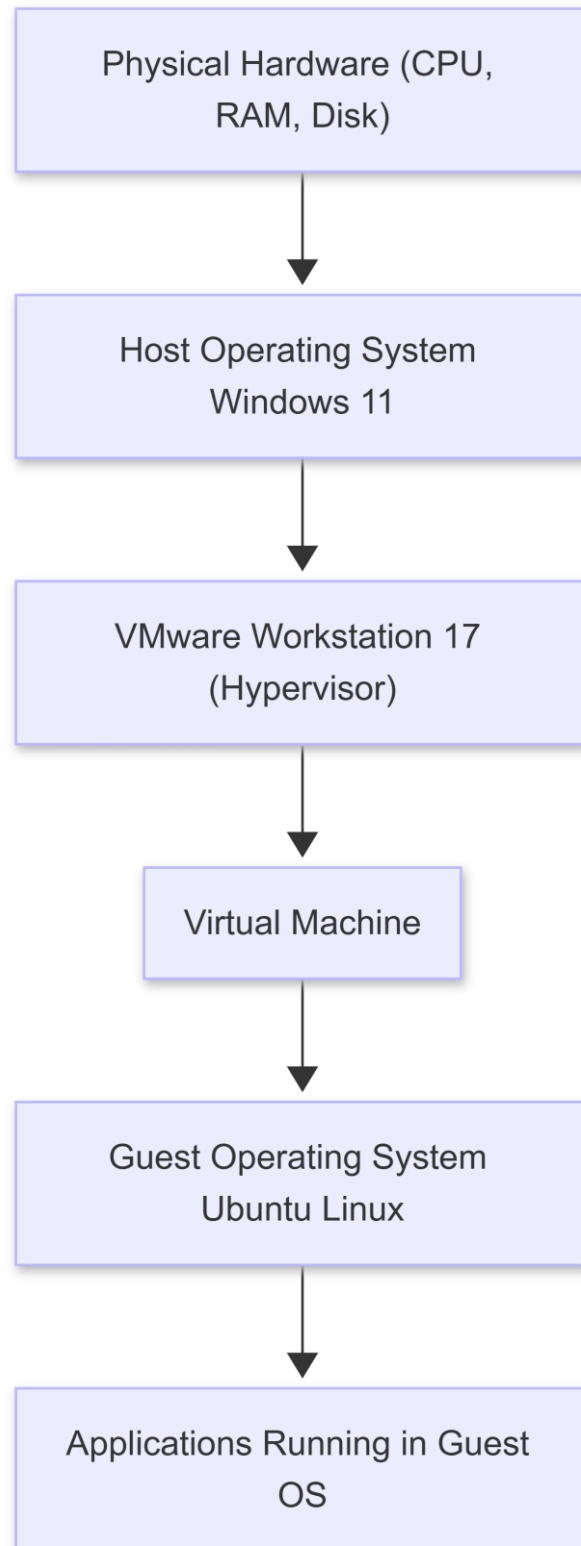
1. To understand the concept of virtualization.
2. To create and configure a Virtual Machine.
3. To install a guest operating system.
4. To understand host and guest OS interaction.
5. To observe how virtualization allows multiple OS environments.

3. Tools and Technologies Used

Component	Description
Host OS	Windows 11
Virtualization Software	VMware Workstation 17
Guest OS	Ubuntu 22.04 LTS
ISO Image	Ubuntu Desktop ISO

4. System Architecture

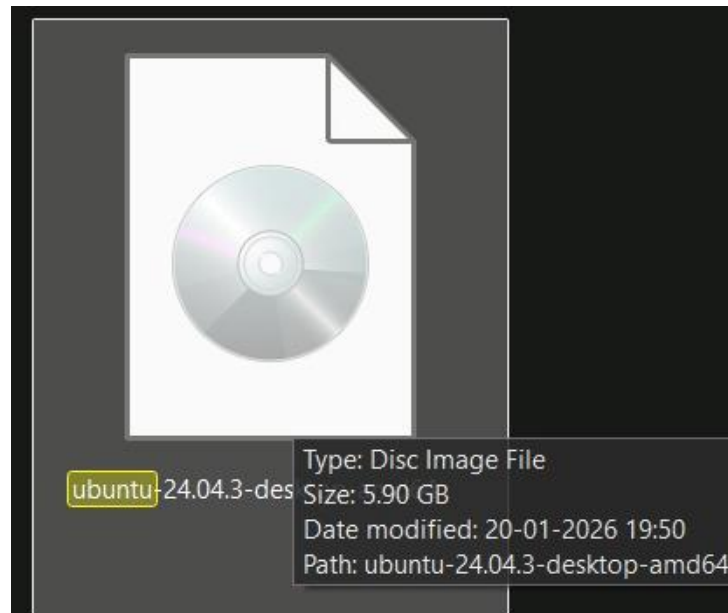
The host system runs VMware Workstation, which creates and manages virtual machines. The Ubuntu OS runs inside a VM as a guest OS.



6. Implementation Details

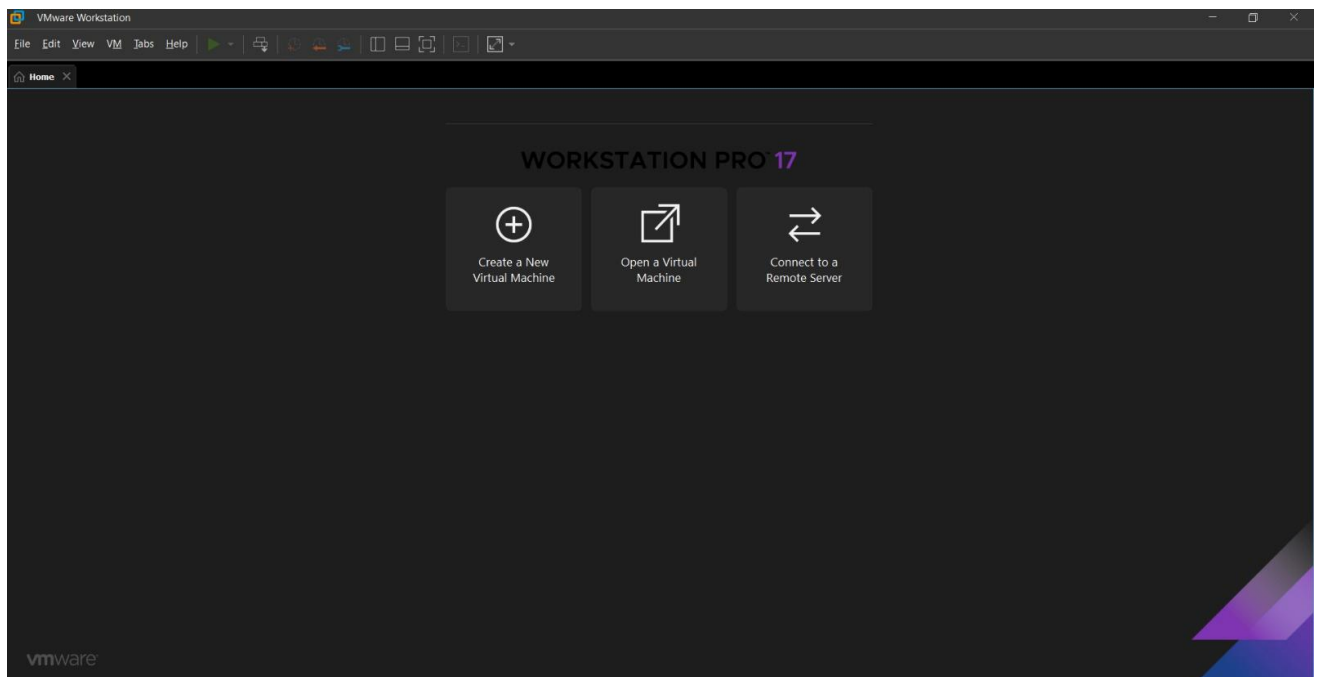
Step 1: Download Ubuntu ISO

Ubuntu 22.04 LTS ISO file was downloaded.



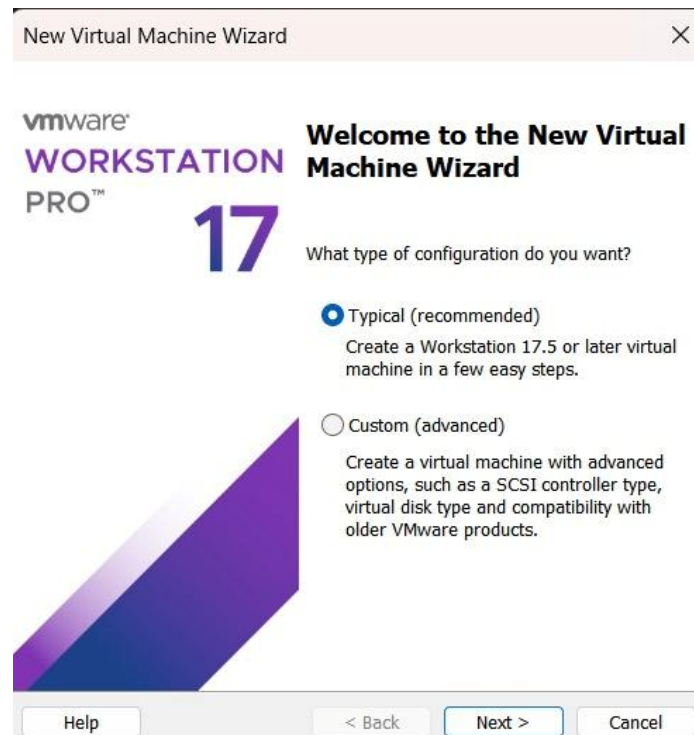
Step 2: Open VMware Workstation

VMware Workstation was launched.



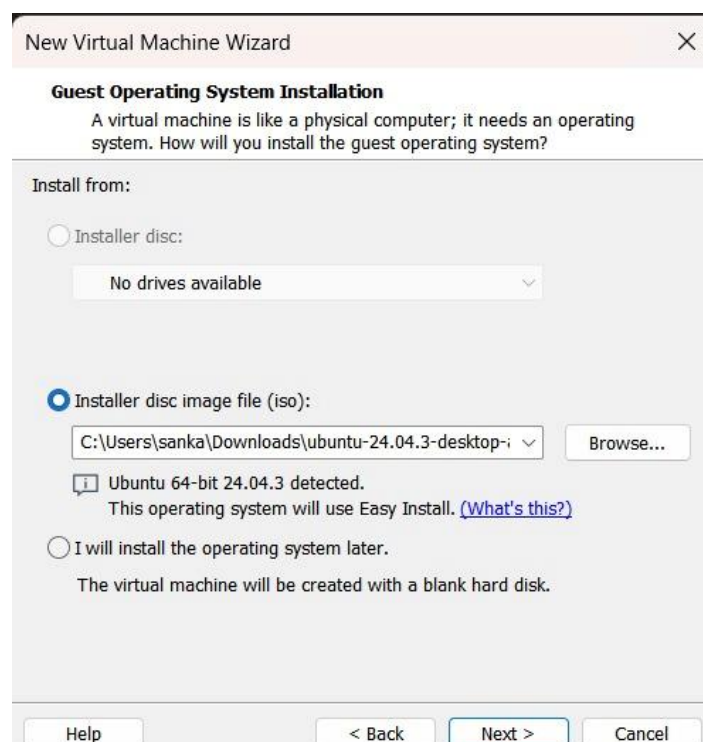
Step 3: Create New Virtual Machine

A new VM was created using the “Typical” configuration.



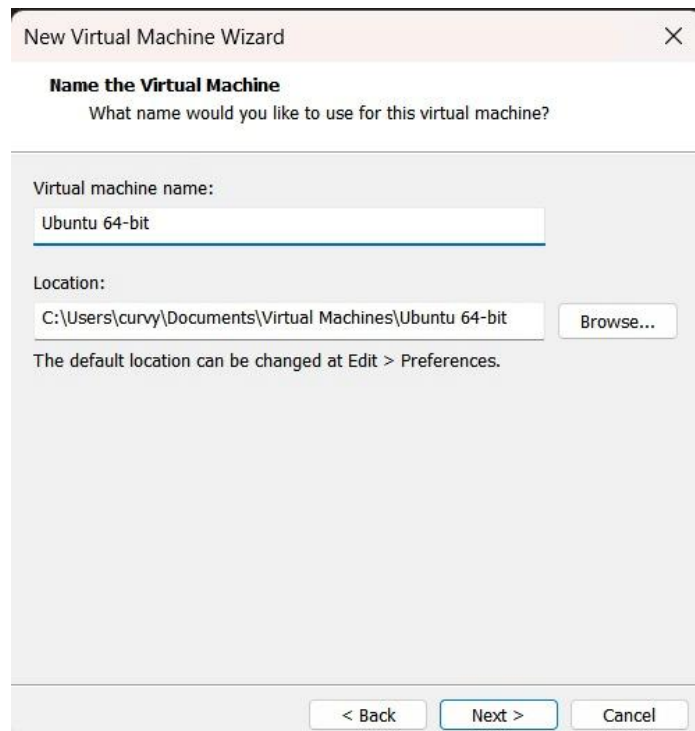
Step 4: Select ISO Image

Ubuntu ISO file was selected as installation media.



Step 5: Configure OS Type

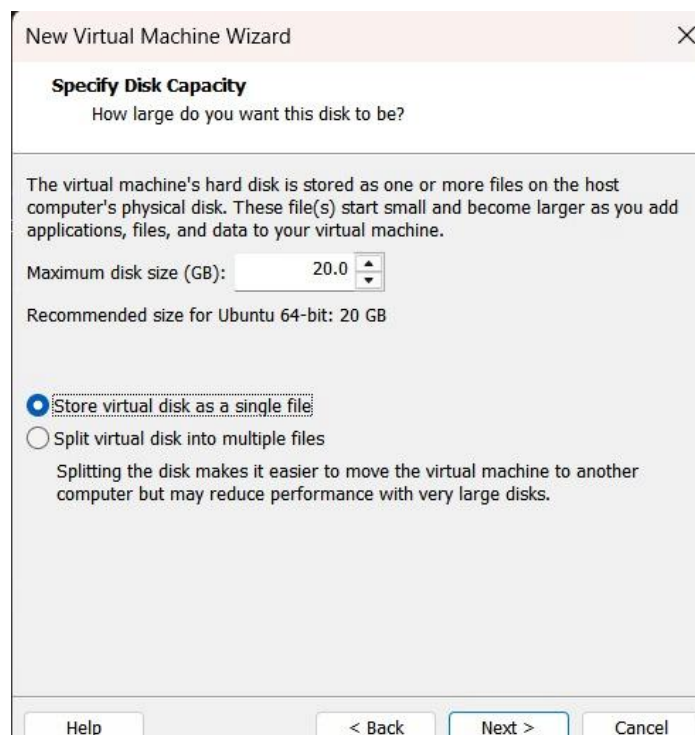
Linux - Ubuntu 64-bit was selected.



The screenshot shows the 'New Virtual Machine Wizard' dialog box. The title bar says 'New Virtual Machine Wizard' with a close button. The main heading is 'Name the Virtual Machine' with the subtext 'What name would you like to use for this virtual machine?'. Below this, there is a text box labeled 'Virtual machine name:' containing 'Ubuntu 64-bit'. Underneath is a text box labeled 'Location:' containing 'C:\Users\curvy\Documents\Virtual Machines\Ubuntu 64-bit' and a 'Browse...' button. A note below the location box states: 'The default location can be changed at Edit > Preferences.' At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'.

Step 6: Allocate Disk Space

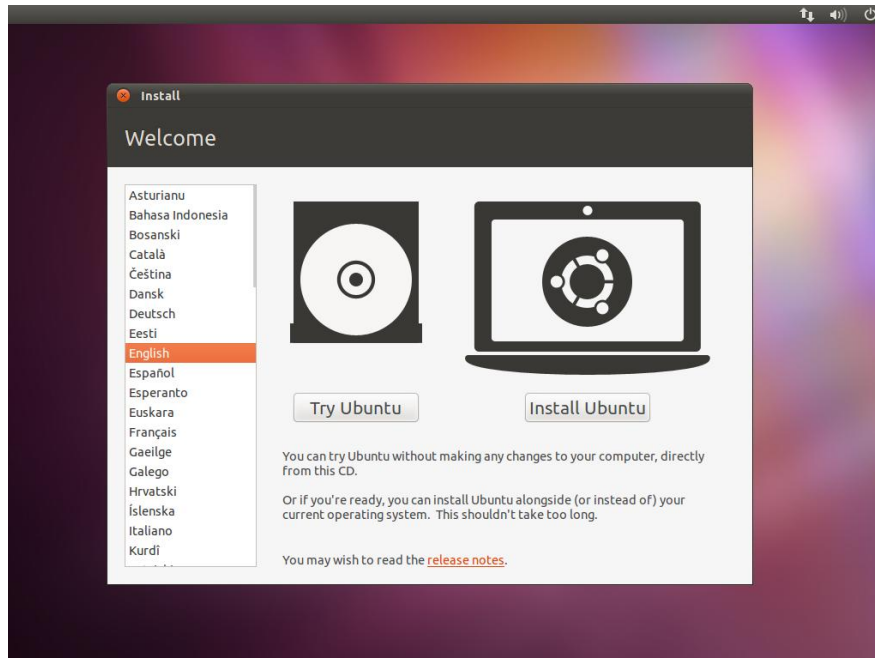
A 20 GB virtual disk was created.



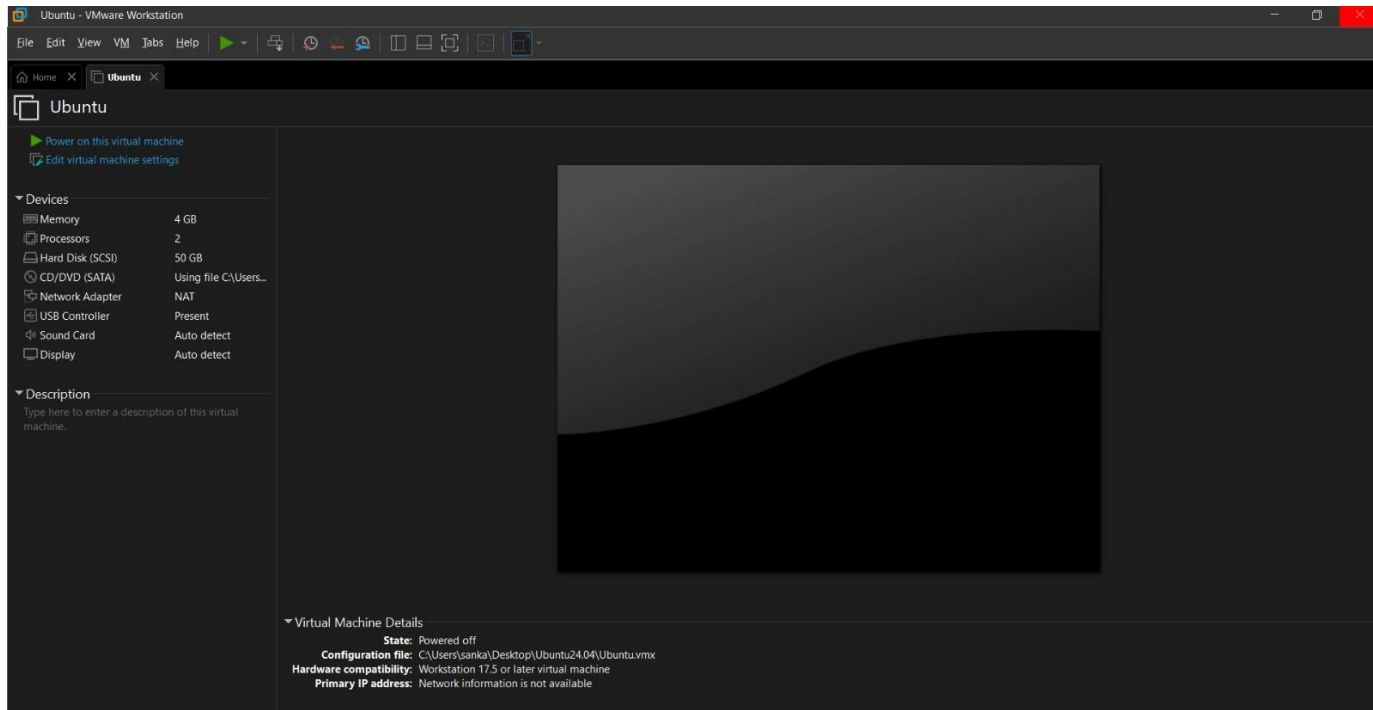
The screenshot shows the 'New Virtual Machine Wizard' dialog box. The title bar says 'New Virtual Machine Wizard' with a close button. The main heading is 'Specify Disk Capacity' with the subtext 'How large do you want this disk to be?'. Below this, there is a paragraph of text: 'The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.' Below the text is a text box labeled 'Maximum disk size (GB):' with '20.0' and a spinner control. Underneath is the text 'Recommended size for Ubuntu 64-bit: 20 GB'. There are two radio button options: 'Store virtual disk as a single file' (which is selected) and 'Split virtual disk into multiple files'. Below the second option is a paragraph of text: 'Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.' At the bottom, there are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'.

Step 7: Install Ubuntu

Ubuntu installation process was started inside the VM.

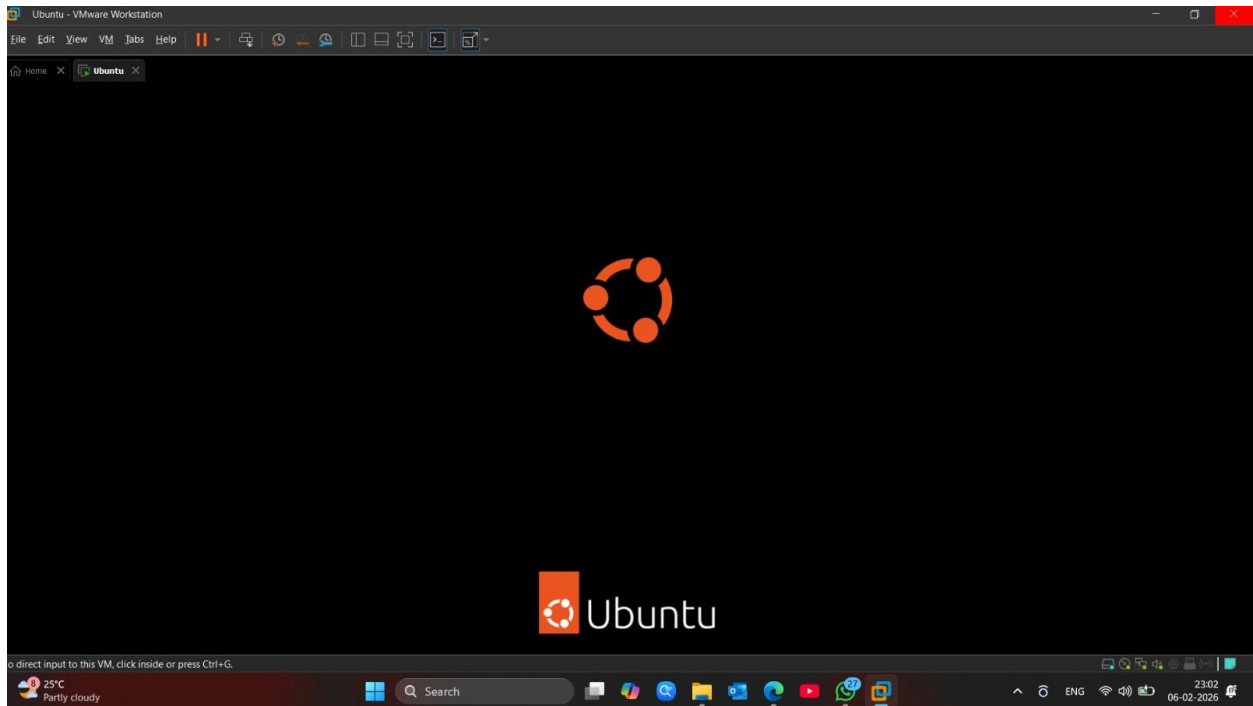


In VMWare:



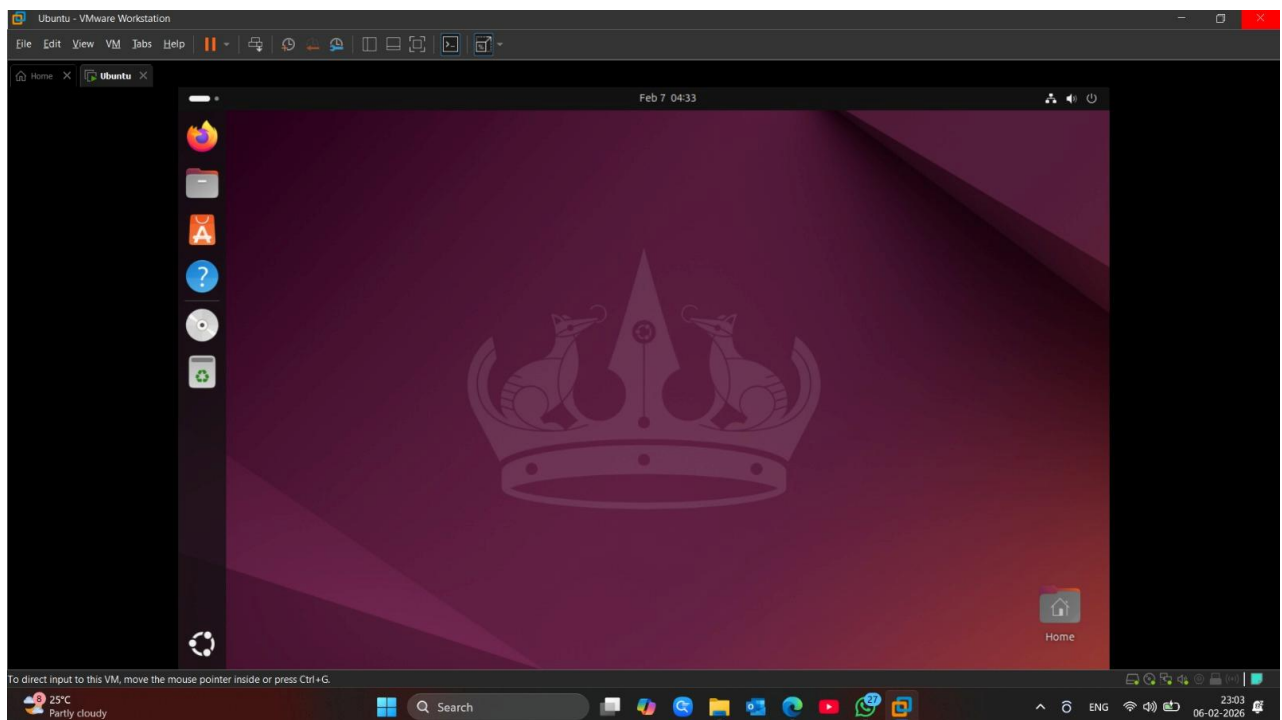
Step 8: Ubuntu Running

After installation, Ubuntu desktop loaded successfully.



9. Output

The Ubuntu operating system ran successfully as a virtual machine inside VMware Workstation on Windows 11.



9. Conclusion

In this experiment, virtualization was successfully implemented using VMware Workstation. A virtual machine running Ubuntu Linux was created and executed inside a Windows host system. This demonstrated how virtualization allows multiple operating systems to run simultaneously on a single physical machine.
